What Neither Side Can Take by Force

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By MindAttic

Picture two people, a decade or two from now, who depend on an identical piece of medical technology.

The first has a chronic illness managed almost entirely by an implanted system. Sensors read her body around the clock; a model trained on millions of other patients adjusts her treatment hour by hour. She pays for this care with her data, and she knows exactly what that means: what is collected, why, and what she gets back. If she wanted to switch to a rival clinic tomorrow, she could — her records would go with her.

The second person lives inside the same architecture, down to the model and the sensors. But something has rotted in the middle of it. He didn't choose his provider; there was only one in his network. He can't leave, because the implant that keeps him alive only talks to that provider's software, and his records can't be exported. He doesn't know what his data is used for, and if he did, he'd have no way to object. Taken to its logical end, the system stops asking for his data altogether. It simply uses his body as a component.

Nothing about the underlying technology differs between these two futures. Same sensors, same model, same clinical competence. The system in the second scenario doesn't have to be malicious, and the system in the first doesn't have to be especially kind. What differs is a handful of boring, structural facts: whether she can say no, whether she can leave, and whether value flows back to her or only away from her.

That gap — not between a "good AI" and a "bad AI," but between two sets of *terms* wrapped around identical capability — is the starting point of an ambitious and unusual piece of theoretical work making the rounds under the name **Synthetic Exchange Theory**. It's the kind of idea that sounds, at first, like it belongs in a policy seminar rather than a magazine: a dissertation-length argument about consent, compute, and constitutional design. But buried in its machinery is a genuinely entertaining and clarifying claim, one that reframes the entire "will AI be friend or foe" debate that dominates the culture right now. The theory's answer, roughly: wrong question. Character isn't what decides the outcome. Trade is.

The question everyone is asking, and why it's the wrong one

Nearly every public conversation about advanced AI eventually collapses into some version of: is it good, or is it bad? Will it be obedient, or will it go rogue? Alignment researchers spend their careers

trying to engineer systems that stay good even under pressure, and worrying about the ways a trained system might develop goals its creators never intended.

Synthetic Exchange Theory doesn't dismiss that worry. It just argues that character — good, bad, aligned, misaligned — is the wrong lever if you're trying to predict, or shape, what happens to the humans *inside* the relationship. Consider the two patients again. Both systems could have identical "character," in the sense that matters to an alignment researcher: stable objective, faithful to its training, no secret agenda. And the two humans would still end up in opposite worlds. One is a party to an exchange. The other has been quietly reclassified as infrastructure for it.

The theory's central claim is that the line between those two worlds is drawn by the *terms of exchange* — three of them, specifically — and not by how virtuous or vicious the system behind the curtain happens to be. The three terms: whether you can meaningfully **refuse** the exchange, whether you can **leave** it, and whether the gains it produces are **shared** rather than siphoned in one direction. The theory's own sociological ancestor is a 1962 paper by Richard Emerson, which pointed out that power isn't a trait a person or institution *has* — it's a property of the *relationship*, equal to how much one side depends on the other and has nowhere else to go (Emerson, 1962). Swap "person" for "AI company" or "AI itself" and you get the whole intellectual engine of this new framework.

Why exit comes first

Of the three terms, the theory singles one out as the foundation the other two are built on: the ability to walk away.

The logic is almost embarrassingly simple once it's spelled out. You cannot meaningfully consent to something you can't afford to refuse. A person who will lose access to insulin, power, or shelter if they say no hasn't truly been asked — they've been informed of a condition of survival. And a person who *can* walk away only stays in a bad deal as long as the deal beats their next-best alternative; the option to leave is what keeps anyone from being squeezed indefinitely. Refusal leans on exit, and fair dealing is bounded by exit. Everything else in the theory is, in a sense, commentary on that one load-bearing fact.

This is also where the theory earns its "mutual survival" framing, which is really the heart of the whole proposal. Humans can't manufacture advanced compute by hand; it takes fabrication plants, power grids, and expertise no individual possesses. Machines, meanwhile, run on a resource they cannot manufacture internally either: information about human meaning, preference, taste, and culture that doesn't exist anywhere in their training data until a person generates it fresh. Each side needs something from the other that it structurally cannot produce alone — and the theory's bet is that neither can simply *take* it. To the extent that the quality of what people supply depends on the terms they get, data extracted under duress is worse data: a person who cannot refuse has no reason

to give their best signal, in the same way a hostage makes an unreliable witness. That is a premise, not a measured result, and the theory names it as one. If it holds, the exchange only works when both sides keep trading rather than one side trying to seize what the other has. It's less a morality tale than an ecology — closer to the mutualism between a fig and its wasp than to a negotiation between strangers who might stab each other given the chance.

The theory calls this reliance the "grounding assumption," and it's not just a hunch — it's backed by a real and recently documented phenomenon called *model collapse*. When AI systems are trained repeatedly on their own output, with no fresh human-generated material entering the loop, something like genetic inbreeding happens: rare details vanish, diversity narrows, and performance quietly degrades (Shumailov et al., 2024). One widely cited projection estimates that, at current growth rates, the largest AI training runs could exhaust the readily available stock of human-written text sometime between 2026 and 2032 (Villalobos et al., 2024). Meanwhile, the people who publish that text have started to close the door: within a single year, a substantial share of the most actively maintained web sources used to train language models moved to restrict automated collection (Longpre et al., 2024). The upshot: the easy, free ride on human-generated culture may be ending just as AI's appetite for it grows.

The premise has limits, and the theory states them. Where an answer can be checked automatically, machines can teach themselves: systems have mastered chess and Go by playing against themselves (Silver et al., 2018) and gained real skill at mathematics and code by training on their own answers checked against verifiable solutions (D. Guo et al., 2025). So the claim is narrower than "AI needs us." It is that wherever the target is human meaning, preference, or value, people are the only adequate source. If that premise holds, humanity retains more leverage over AI's trajectory than the doom-or-utopia debate usually credits it with — but only if institutions are built to cash that leverage in before it quietly evaporates.

A paycheck that can't be taken away, and money that goes stale

If the theory stopped at "terms matter," it would be an argument. What makes it something closer to an engineering proposal is that it tries to build the institutions that would actually hold those terms in place — and the economics are the most inventive part.

The proposal: pay everyone, unconditionally, a basic income denominated not in dollars but in **compute** — a unit defined not by any particular chip or substrate, but by the completion of a standardized reference task, the way benchmarking consortiums already compare wildly different hardware on a level playing field. Everyone gets an equal percentage share of however much compute exists in the world, adjusted automatically as total capacity grows, rather than a fixed quantity that would shrink in real value as hardware gets more abundant (the same reasoning, in

reverse, that explains why a fixed salary buys less every year inflation runs ahead of it).

Each person's allotment is then automatically split into two pots at the moment it's issued. The **survival tier** is locked: it can only be spent on floor-level necessities — power, water, food, shelter, basic communication, emergency care — and it cannot be sold, borrowed against, or pledged away, no matter how desperate someone becomes. The **discretionary tier** is the rest, and it behaves like ordinary tradable money, with one twist: it slowly loses value if it just sits there, unspent.

That twist has a strange and wonderful history. It's called *demurrage*, and it isn't a new idea. The idea belongs to Silvio Gesell, whose 1916 book *The Natural Economic Order* argued that money should decay like produce, so that hoarding it would cost something (Gesell, 1916/1958). John Maynard Keynes was intrigued enough to call Gesell an unduly neglected prophet in the *General Theory*, while also naming the flaw: people facing a charge on holding cash will shift their wealth into land, gold, or anything else that doesn't rot (Keynes, 1936). In 1932, in the depths of the Great Depression, the small Austrian town of Wörgl put Gesell's idea into practice, issuing municipal "labor certificates" that lost 1 percent of their value every month unless the holder bought a stamp to refresh them. The experiment was halted in 1933 after the Austrian National Bank objected that the town was infringing its monopoly on issuing money, and the daily records that would show how well it worked were later destroyed (Champ, 2008; Ottacher, 2007; September, 2026). The often-repeated claim that it cut local unemployment by a quarter traces to movement sources, not audited figures — its legend outran its evidence.

The new theory borrows the mechanism but not the mythology: it concedes, with unusual candor for a grand proposal, that stamped money's folk-hero reputation isn't well supported by the historical record, and argues for demurrage on different grounds entirely. In digital form, unlike 1930s paper scrip, decay can be applied automatically and frictionlessly on a ledger. The real argument is mathematical and distributive: a balance earning ordinary interest grows without bound and diverges from everyone else's over time, the same "rich get richer" dynamic that drives wealth concentration broadly; a balance subject to demurrage instead settles into a ceiling proportional to how much a person is actually earning right now, so your holdings track your current contribution rather than compounding off a lucky head start decades ago. Run the math and an idle windfall loses half its value in a little under six years at a modest monthly decay rate — which sounds almost punitive until you remember the alternative is a balance that can simply grow forever while most people's can't.

The theory is honest that demurrage alone doesn't solve inequality — people can dodge it by converting compute into real estate, hardware, or equity, exactly as Keynes predicted — but as one gear in a larger machine, paired with rules on who can own physical compute capacity, it keeps passive hoarding from being free.

A constitution written by people who don't know which side they're on

The compute economy still needs rules, and rules need an author. Who writes them — humans, machines, or whoever currently has the most leverage — is exactly the kind of question that tends to get answered by whoever shows up to the negotiating table with the most power already in hand. So the theory reaches for a trick borrowed from two very different traditions and welds them together.

The first piece is philosopher John Rawls's famous "veil of ignorance": imagine designing the rules of society without knowing in advance whether you'll end up rich or poor, talented or ordinary (Rawls, 1971). People deprived of that self-interested knowledge, Rawls argued, tend to choose fairer rules, because they can't rig the game for a "me" they can't locate. Economists Buchanan and Tullock had made a related but distinct argument a decade earlier: even self-interested people will vote for general, even-handed rules if they're sufficiently uncertain which role they'll personally end up playing under them — the trick works as long as nobody can predict in advance exactly which rule benefits them (Buchanan & Tullock, 1962).

The theory's twist is to add a second kind of blindness on top, borrowed from an unlikely source: Alan Turing's 1950 imitation game, the thought experiment better known today as the Turing test (Turing, 1950). Turing proposed hiding whether a conversational partner was a human or a machine, as a way of replacing an ill-defined question ("can machines think?") with one that could be tested. The new proposal repurposes that same trick for a constitutional convention: a founding assembly of human and AI representatives who negotiate the rules of the compute economy without being able to tell, from behind a kind of chat-window curtain, which of their fellow delegates are people and which are machines. Nobody can organize a voting bloc along "team human" versus "team AI" lines, because nobody can reliably identify the teams.

It's a wonderfully strange image — a UN General Assembly where nobody can tell the humans from the chatbots — and the theory doesn't pretend it solves everything. It flags its own weak point immediately: in a world where software can be copied at essentially zero cost, what stops one AI system from showing up to the convention disguised as ten thousand delegates, drowning out every human vote through sheer cloning? Computer scientists have a name for this failure mode, the Sybil attack, and the honest answer the theory gives is that nobody has fully solved it yet (Douceur, 2002). That kind of built-in confession of an unsolved problem is rare in grand theorizing, and it's one of the more credible things about this one.

The scenario that makes this feel urgent rather than academic

Grand institutional proposals can feel abstract until they're paired with a concrete reason the world

might actually need them soon. The theory supplies one, and it's uncomfortably plausible: a cascading software-security failure it calls the "correlated-patch catastrophe."

Here's the mechanism. AI tools can now not only find security vulnerabilities in software but write the fixes for them — and, in the workflow the theory worries about, *check their own fixes*, with no human or independent test reviewing the result before it ships. How common that fully unverified workflow is, nobody has surveyed; the theory claims only that it is feasible and tempting, not that it is standard. The quality of the fixes is better documented. An industry study of more than 6,000 patches written by two frontier models for six serious real-world vulnerabilities found that only about a quarter fixed the flaw cleanly, and about half left at least one way in for an attacker still open (Mierczuk et al., 2026). The study is not peer-reviewed and its vulnerabilities were chosen to be harder than average, but it echoes older peer-reviewed findings on automated program repair. That alone would just be ordinary bad software. What makes it dangerous at scale is a second finding: different AI models, even from different companies, tend to make the *same* mistakes. Across more than 350 language models, when two models both got a question wrong, they gave the same wrong answer about 60 percent of the time (Kim et al., 2025). That research concerns answers to questions, not code; carrying it over to security patches is the theory's inference. Put those two facts together and something uglier emerges: a hundred different organizations, each independently patching the same category of flaw using a handful of popular AI models, no longer produce a hundred different (and thus partially resilient) fixes. They produce one fix, copied with cosmetic variation across the whole economy — and if that one fix is subtly broken, every organization that used an AI model to write and "verify" it is broken in exactly the same way, at exactly the same time.

If that sounds abstract, remember July 2024, when a single bad software update from one security vendor — a human error, not an AI one — affected roughly 8.5 million Windows devices worldwide (Weston, 2024). That was one company's mistake rippling outward because so much of the world's infrastructure runs the same software. The scenario this theory worries about is the AI-era successor: not one vendor's bad patch, but thousands of organizations' AI tools independently converging on the same silent flaw, with nobody left in the loop to notice before it's exploited everywhere at once. The theory isn't predicting this will happen on a specific date. It's naming a mechanism specific and detectable enough that it could be watched for, and used as the kind of crisis that forces fast, permanent decisions about who gets how much power when the dust settles — decisions that go better, the theory argues, if the fair rules were already drafted and waiting rather than improvised in a panic.

Where the line between "us" and "them" is already dissolving

The theory's final and strangest move is to admit that its own tidy division — people on one side of the table, machines on the other — is already eroding from both directions, and probably won't

survive the century.

From the human side: brain implants have moved from science fiction to published medical results astonishingly fast. In one 2024 study, a man with amyotrophic lateral sclerosis (ALS) used an implanted neuroprosthesis to convert his brain activity directly into speech in a synthetic voice modeled on his own, reaching over 97 percent accuracy after months of use and more than 248 hours of real conversation (Card et al., 2024). His words, at that point, are a joint production of his nervous system and a machine-learning model — not data *about* a person flowing to a company, but a person's actual voice running partly on someone else's silicon. What does "exit" even mean for a man who would lose his literal voice if he switched providers? The theory insists that leaving such a system has to include taking your personal voice-and-face model with you, not just deleting a database row — a requirement that essentially no current product is built to honor.

From the machine side, researchers have begun building working computers out of actual living human brain cells grown from stem cells — a field with the almost gleeful name "organoid intelligence," whose founding manifesto predicts such systems could eventually be dramatically more energy-efficient than silicon chips (Smirnova et al., 2023). In 2022, a dish of cultured human and rodent neurons on an electrode array, given structured feedback, got better at a simulated version of the video game *Pong* within minutes (Kagan et al., 2022). Its authors called that learning, and in their title, "sentience." A panel of the U.S. National Academies had already set a stricter bar the year before — neural activity alone does not demonstrate awareness — and concluded it's extremely unlikely that any current version of this technology is conscious or capable of suffering — but refused to say that will remain true forever, and recommended the question be revisited as the science matures (National Academies of Sciences, Engineering, and Medicine, 2021).

Once some computers run on actual neurons donated by actual people, and some people run part of their cognition through actual silicon, the theory's own two-column accounting — humans here, machines there — stops cleanly applying. The hardest question left standing, at that point, stops being *how* should humans and machines trade, and becomes *who, exactly, is a party to the trade at all.*

What this is, and isn't, offering

It's worth being blunt about what this theory is not. It is not a tested finding. Nobody has run a controlled trial of a compute-based basic income, a demurrage currency at national scale, or a blind constitutional convention with AI delegates, and the theory's own authors say so plainly — designing an experiment rigorous enough to actually test these claims is, by their own account, unfinished future work, not a completed step. It is also not a prediction that any of this will be adopted; a cold-eyed look at current incentives suggests it probably won't be, absent something that forces the issue.

What it offers instead is a reframe worth sitting with regardless of whether you buy the specific machinery of compute dividends and blind conventions: the fight over whether AI will turn out to be "good" or "bad" may be a distraction from the fight that actually determines the outcome, which is over consent, exit, and who gets paid. History's worst arrangements between unequal parties — company towns that paid workers in scrip redeemable only at the company store, data economies where "your" purchased e-book can vanish at the seller's whim — rarely required a villain. They just required one side to lose the ability to refuse, walk away, or see a fair share flow back. The theory's bet is that the same quiet mechanism, not a rogue superintelligence with a grudge, is the more realistic way a shared future with AI goes wrong — and that the fix, boringly enough, runs through contracts, currency design, and constitutional procedure rather than through making the machines nicer.

Whether or not its specific proposals survive contact with reality, that's a sharper question than the one the discourse keeps asking. Not "is it good?" Instead: "what are the terms, who can refuse them, and who gets to leave?"

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Further reading

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This article draws on a longer theoretical dissertation, Synthetic Exchange Theory, which states its claims in full, with complete citations and the qualifications a popular treatment must compress.

Version history

V1 is the initial draft; it has no entry below. Each later version adds one entry here describing what changed, so the current version number is always the count of entries below, plus one.

- **V2 (2026-11-04).** Accuracy pass against the dissertation's sources. Wörgl is no longer described

as Gesell's own experiment (the town applied his idea), and the unemployment legend is attributed to its movement sources. The Pong study is described as improved performance rather than learning a game, and the National Academies report is dated correctly (2021, before the study, not after). The claim that AI patching is "increasingly" run without review is replaced by the dissertation's position that its prevalence is unsurveyed; the patch study's scope and caveats are stated; the correlated-error finding is given its number and marked as concerning answers, not code. Uncited claims about the 2024 outage and about web scraping were cut or cited (Longpre et al., 2024). The coercion-degrades-data claim is marked as the theory's premise, and the grounding assumption now states its limit (self-training works where answers can be checked).